

WASP-43b

R-0591

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_240229 · created 2026-07-24T23:12:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460369.8537 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.057
Transit depth (Rp/Rs)^2	3.8 +/- 2.22 [%]
Orbital Inclination (inc)	79.06 +/- 2.87
Airmass coefficient 1 (a1)	0.98 +/- 0.1
Airmass coefficient 2 (a2)	0.03 +/- 0.11
Scatter in the residuals of the lightcurve fit is	10.52 %
Best Comparison Star	#3 - [399, 63]
Optimal Aperture	2.69
Optimal Annulus	5.92
Transit Duration (day)	0.036 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.057 vs 0.1594 (deviation 0.37σ)
Duration fit vs published	0.036 d vs 0.0512 d (deviation 0.5σ)
Mid-time O–C	1.2 min
Depth SNR	2.0
Residual scatter	10.52 %

Light curve

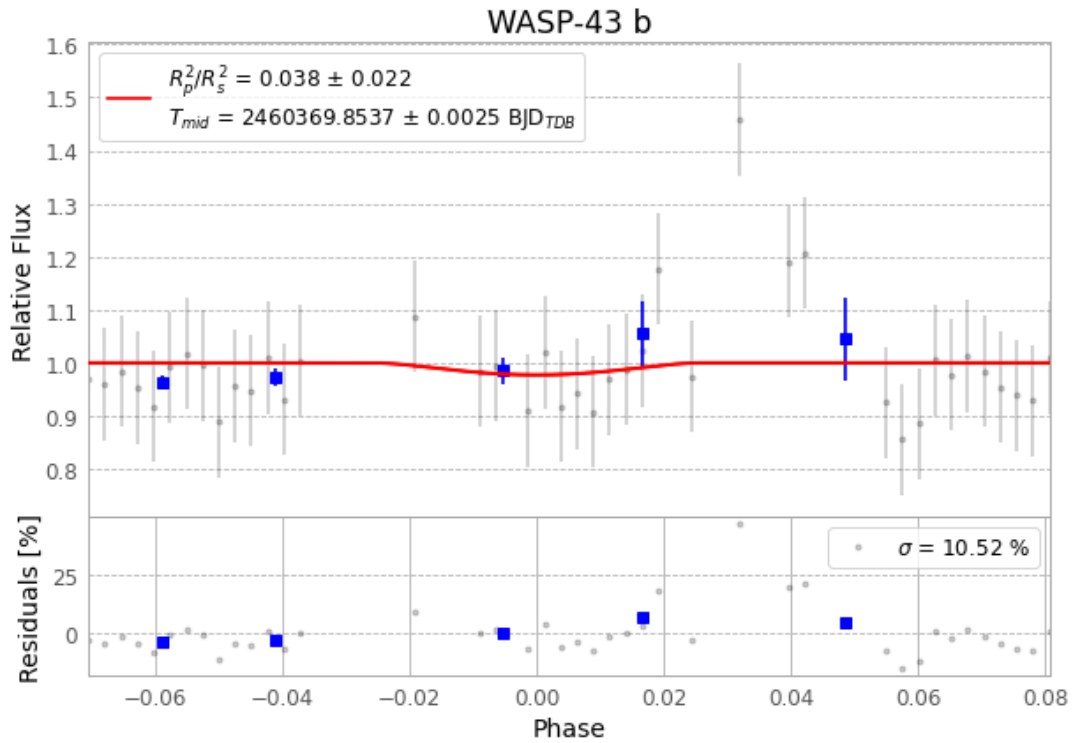


Figure 1 — FinalLightCurve_WASP-43 b_2024-02-28.png (SHA-256 bafeb6cf5b9279...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [486, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 587ea669b8c23282...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

60 FITS frames (39 MB), WASP-43240229065715.FITS → WASP-43240229095415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240229.log (2342fe3232db...), FinalLightCurve_WASP-43 b_2024-02-28.png (baf5b6cf5b9279...), FinalLightCurve_WASP-43 b_2024-02-28.csv (1177a559d3c1...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 23:12Z.